Management Information Systems

Week 3

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Last Week…

- Information systems are a core driver of modern business transformation, enabling efficiency, coordination, decision-making, and ***competitive advantage***

- An information system is composed of ***management, organizational, and technological*** components that work together to support business processes and decisions.

- Technology alone does not create value; _____ are essential for information systems to deliver real business benefits.

Computers vs. Information System

- **Computer-based information systems use computer technology to transform data into information**
  - Computers and programs are the basic ***building blocks*** of information systems but don't create specific organizational information by themselves.
- **One must comprehend the problems they address, their design elements, and the organizational processes that lead to these solutions.**

A house is an appropriate analogy. Houses are built with hammers, nails, and wood, but these do not make a house. The architecture, design, setting, landscaping, and all of the decisions that lead to the creation of these features are part of the house and are crucial for solving the problem of putting a roof over one's head. Computers and programs are the hammers, nails, and lumber of computer-based information systems, but alone they cannot produce the information a particular organization needs. To understand information systems, you must understand the problems they are designed to solve, their architectural and design elements, and the organizational processes that lead to the solutions.

Image Source: Berkshire Hathaway, Eden Builders; Your Own Architect

Information Technology vs. Information Systems

- **Computers and computer programs are the building blocks of information systems**

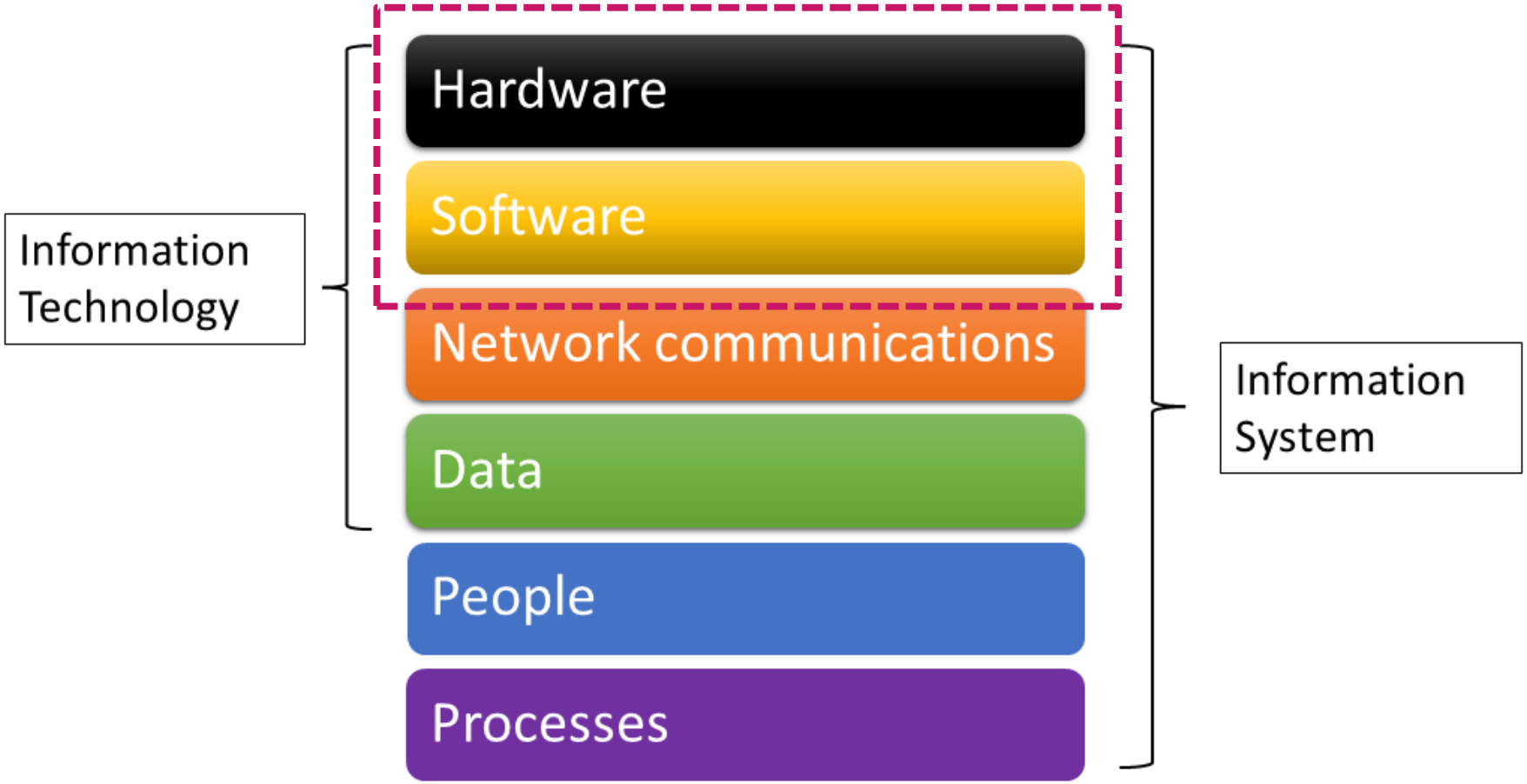
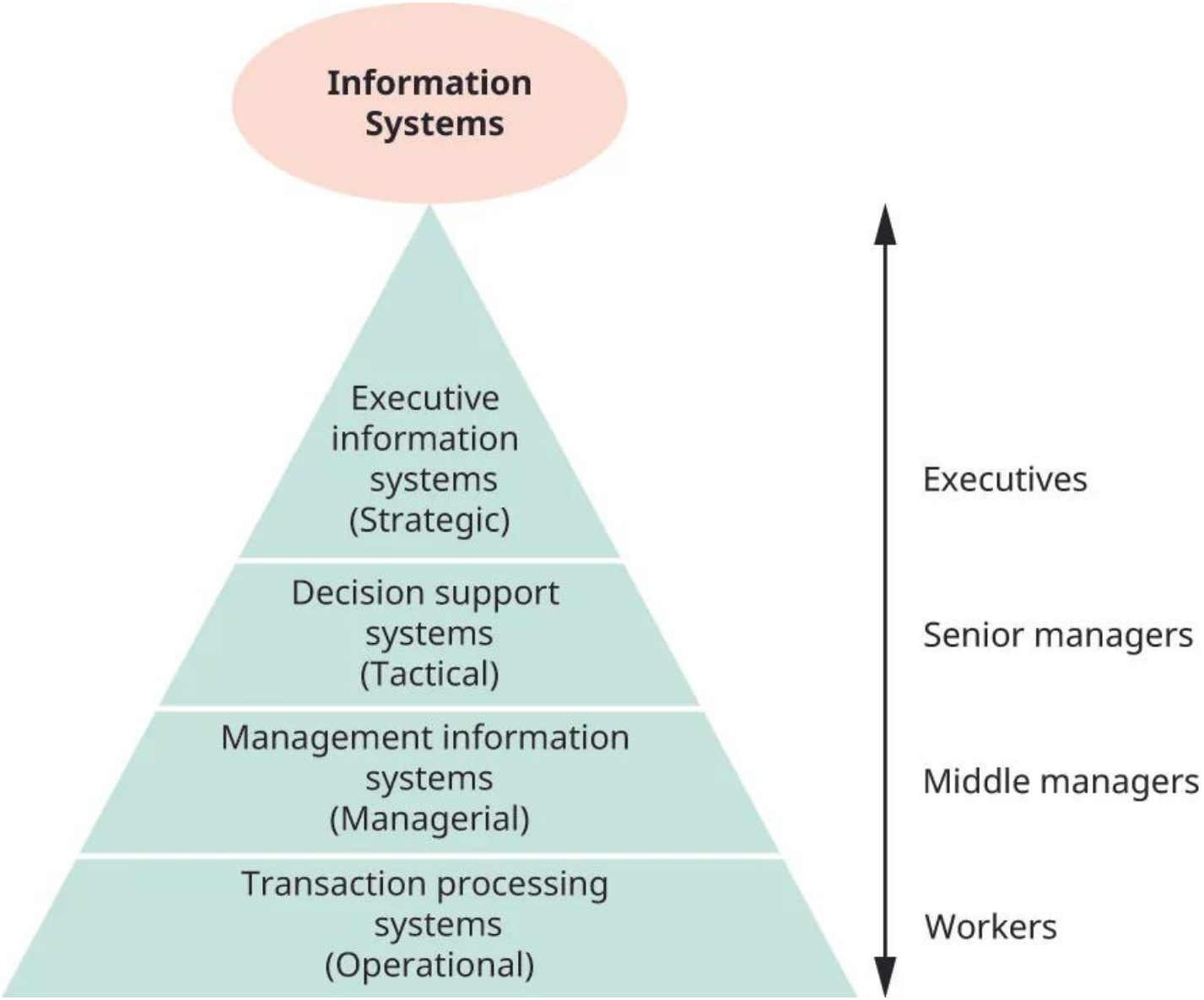


Image Source: Ly-Huong T. Pham and Tejal Desai-Naik

MIS vs. IS



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Weekly Topics

- **Subject to change without prior notice**

W	Ch	Topic	W	Ch	Topic
1		Orientation Introduction to MIS	9	7	Telecommunications, the internet, and wireless technology
2	1	Information Systems in Global business today	10	8	Securing information systems
3	**2**	**Global E-business and Collaboration**	11		*No Class*
4	3	Information Systems, Organizations, and strategy	12	9	Achieving operational excellence and customer intimacy
5	4	Ethical and Social issues in information systems	13	10	E-commerce: digital markets, digital goods
6	5	IT infrastructure and emerging technologies	14	11	Managing knowledge
7	6	Foundations of Business Intelligence	15	12/ 13	Enhancing decision making / Building information systems
8		*Midterm Exam*	*16*		*Final Exam*

Week 3

Global E-business and Collaboration

Questions to Answer

1. What are ***business processes***? How are they related to information systems?

2. How do systems serve the ***different management groups*** in a business, and how do systems that link the enterprise improve organizational performance?

3. Why are systems for ***collaboration and social business*** so important, and what technologies do they use?

4. What is the ***role*** of the information systems function in a business?

2.1
What are business processes?
&
How are they related to information systems?

Why Businesses Need Information Systems

- **To operate, businesses must deal with many different pieces of information about suppliers, customers, employees, invoices, payments, and their products and services.**

- **They must *organize* work activities that use this information to operate *efficiently* and enhance the firm's overall performance.**

- **Information systems allow firms to *manage* all their information, make better decisions, and improve the *execution of their business processes*.**

Business Processes

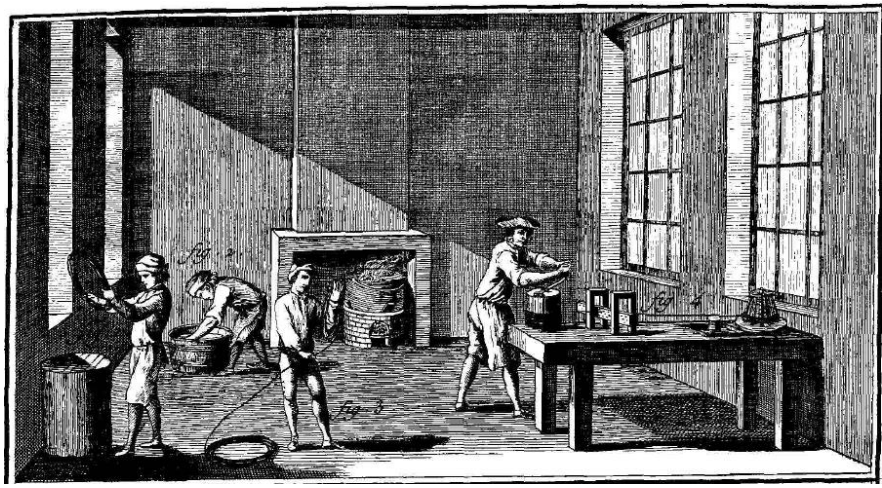
- **Business processes: *the collection of activities* that are required to produce a product or service**
  - These activities are supported by ***material, information, and knowledge flows*** among the participants in business processes.
  - Business processes also refer to the ***unique ways*** in which organizations coordinate work, information, and knowledge
  - Firm performance relies heavily on the design and coordination of its business processes.
    - Well-designed business processes can provide a competitive advantage by facilitating innovation and superior execution compared to rivals.
    - Conversely, inefficient business processes can hinder organizational responsiveness and efficiency, potentially becoming liabilities.

Adam Smith's Pin-Maker Example

- **Division of Labor**

- **Factory #1**
  - Work separately; each employee makes a pin
  - 10 employees make 200 pins per day

- **Factory #2**
  - OP1: draws out the wire
  - OP2: straight the wire
  - OP3: cut the wire
  - OP4: points it
  - OP5: Grinds it at the top to receive the head
  - …
  - (18 operations)



  - 10 employees make ?? pins per day

Business Processes

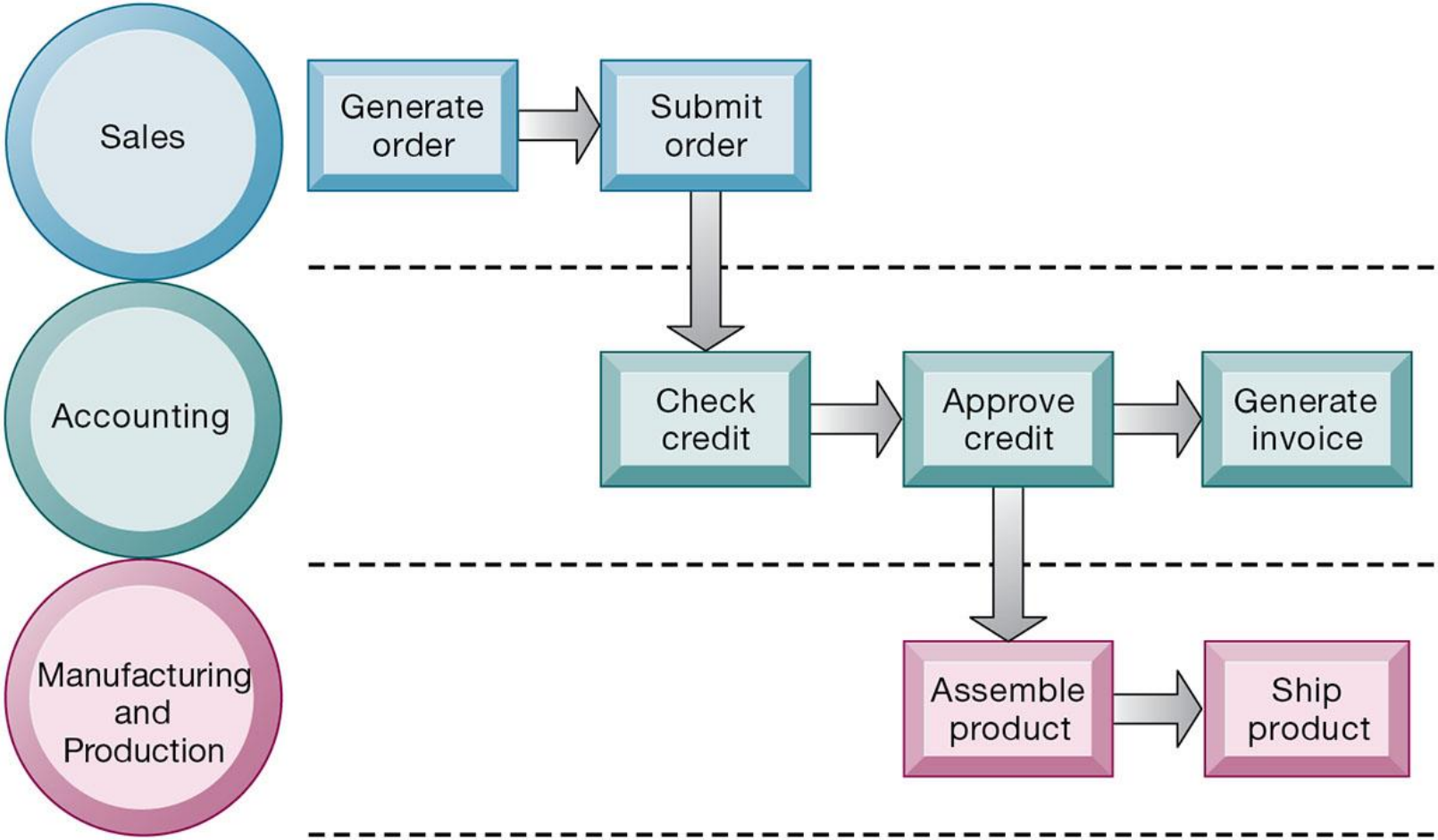
- **Every business can be seen as a collection of business processes, some of which are part of more extensive encompassing processes.**

TABLE 2.1 EXAMPLES OF FUNCTIONAL BUSINESS PROCESSES

FUNCTIONAL AREA	BUSINESS PROCESS
Manufacturing and production	Assembling the product
	Checking for quality
	Producing bills of materials
Sales and marketing	Identifying customers
	Making customers aware of the product
	Selling the product
Finance and accounting	Paying creditors
	Creating financial statements
	Managing cash accounts
Human resources	Hiring employees
	Evaluating employees' job performance
	Enrolling employees in benefits plans

Business Processes

- **It may be tied to a functional area or be cross-functional**
  - Cross-functional: business processes cross ***many different functional areas*** and require coordination across departments.



Example

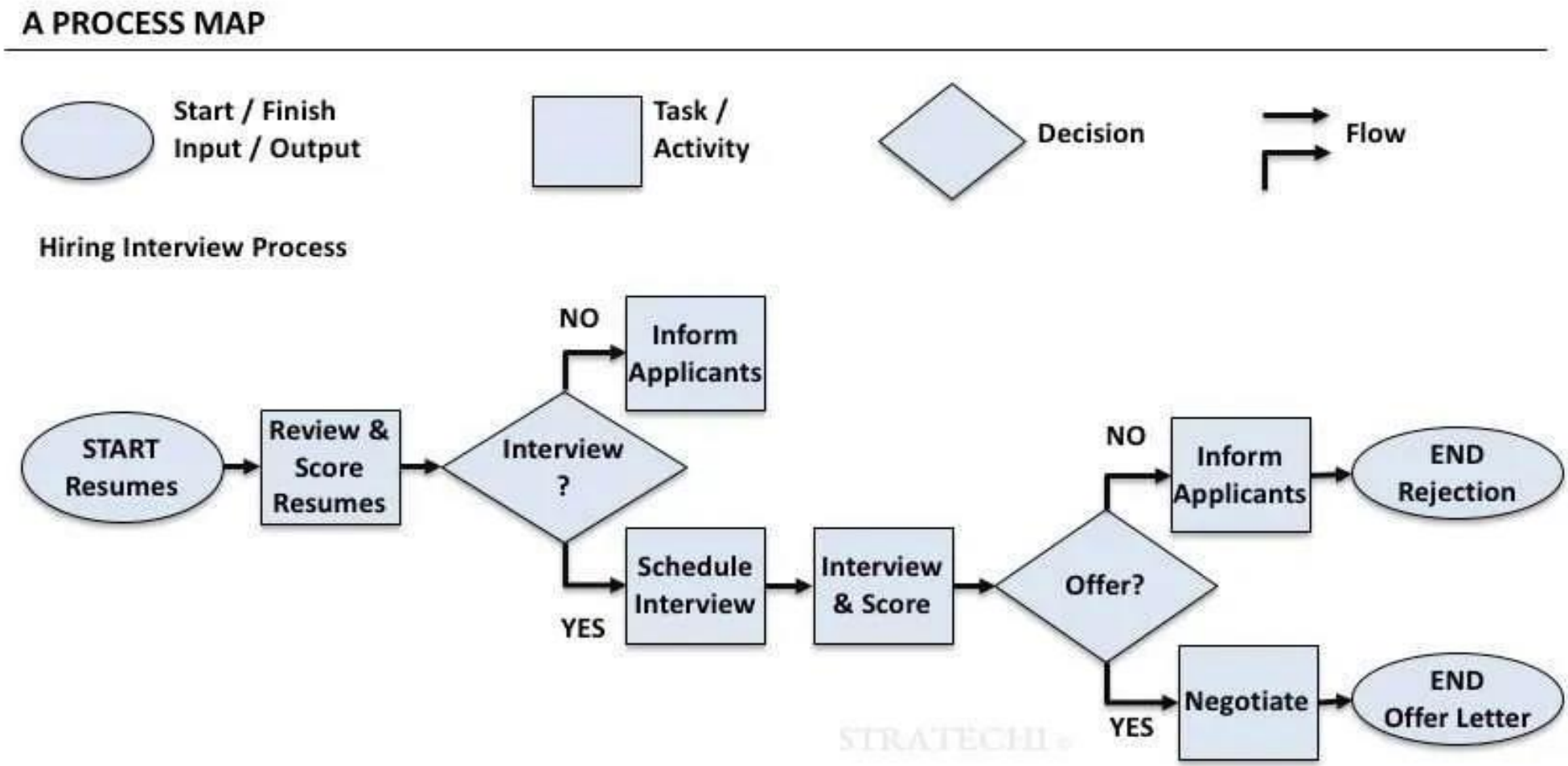


Image Source: Stratechi.com

How IT Improves Business Processes

- **Increasing efficiency of *existing* processes**
  - ***Automation of Manual Tasks***: Initial improvements involved automating manual tasks, like credit checks or generating invoices and shipping orders

- **Enabling entirely *new* processes**
  - IS changes how ***information flows*** within a business, allowing more comprehensive access and sharing of information
  - IS enables tasks that used to be sequential to be ***conducted simultaneously***, reducing process time and increasing efficiency.
  - IS can ***eliminate delays in decision-making*** processes, leading to faster responses to business needs.
  - IS facilitates ***entirely new business models*** such as digital downloads, online shopping, and streaming services.

Business Process Analysis

- **Importance of Business Process Analysis:**
  - Understanding and analyzing business processes is crucial for recognizing how a business operates.
  - This analysis can identify opportunities to improve efficiency and effectiveness through information technology.

- **Examines business processes with a view to understanding how they might be improved by *using IT* to achieve greater efficiency, innovation, and customer service.**

Discussion

- **Which process in your daily life do you find most problematic, and why? Would IT help in simplifying the process? Discuss with your partner.**

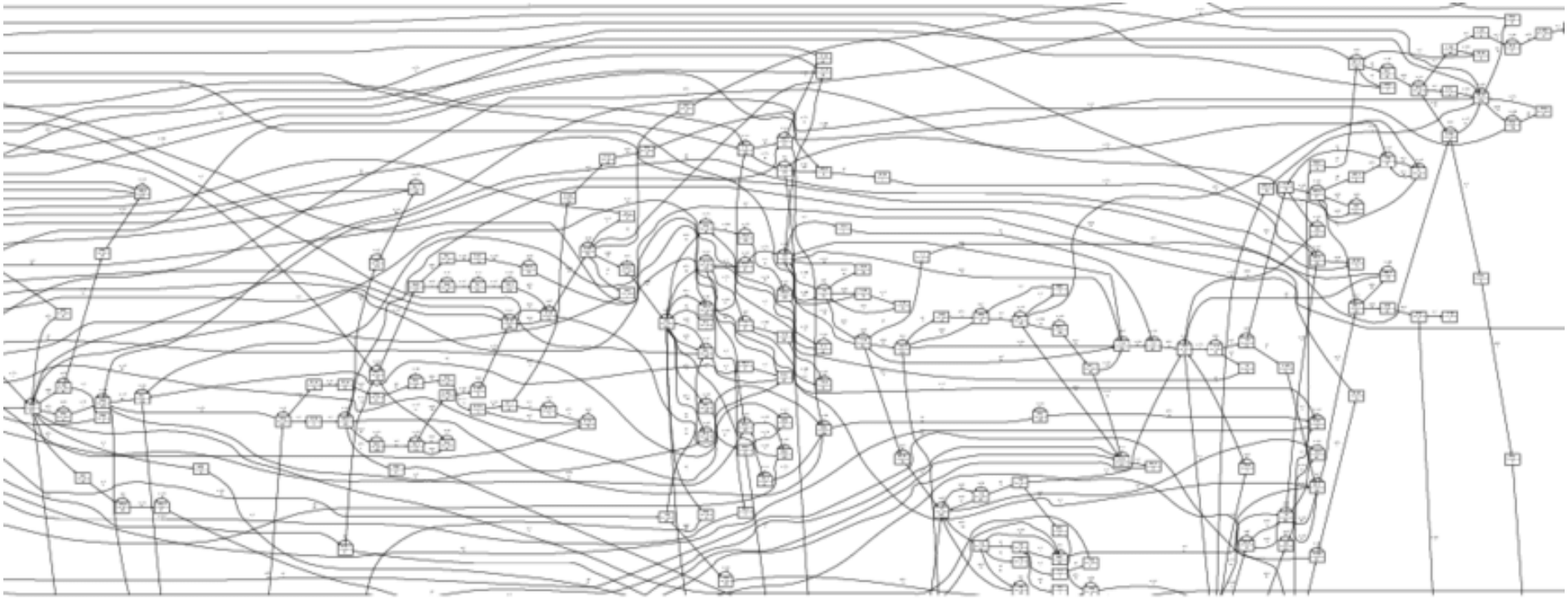


Image Source: Gunther and Aalst (2007). Fuzzy Mining – Adaptive Process Simplification Based on Multi-perspective Metrics

2.2

How do systems serve the different management groups in a business?

&

How do systems that link the enterprise improve organizational performance?

Systems for Different Management Groups

- **Different organizational interests, specialties, and levels necessitate *varied information systems***
  - Since no single system can fulfill all informational requirements

- **Emphasizes the shift from independent systems for specific functions to *comprehensive, integrated solutions* that facilitate broader organizational objectives and decision-making processes.**

- **Different management levels require distinct systems tailored to their decision-making needs.**

Systems for Different Management Groups

- **TPS, MIS, DSS, ESS**

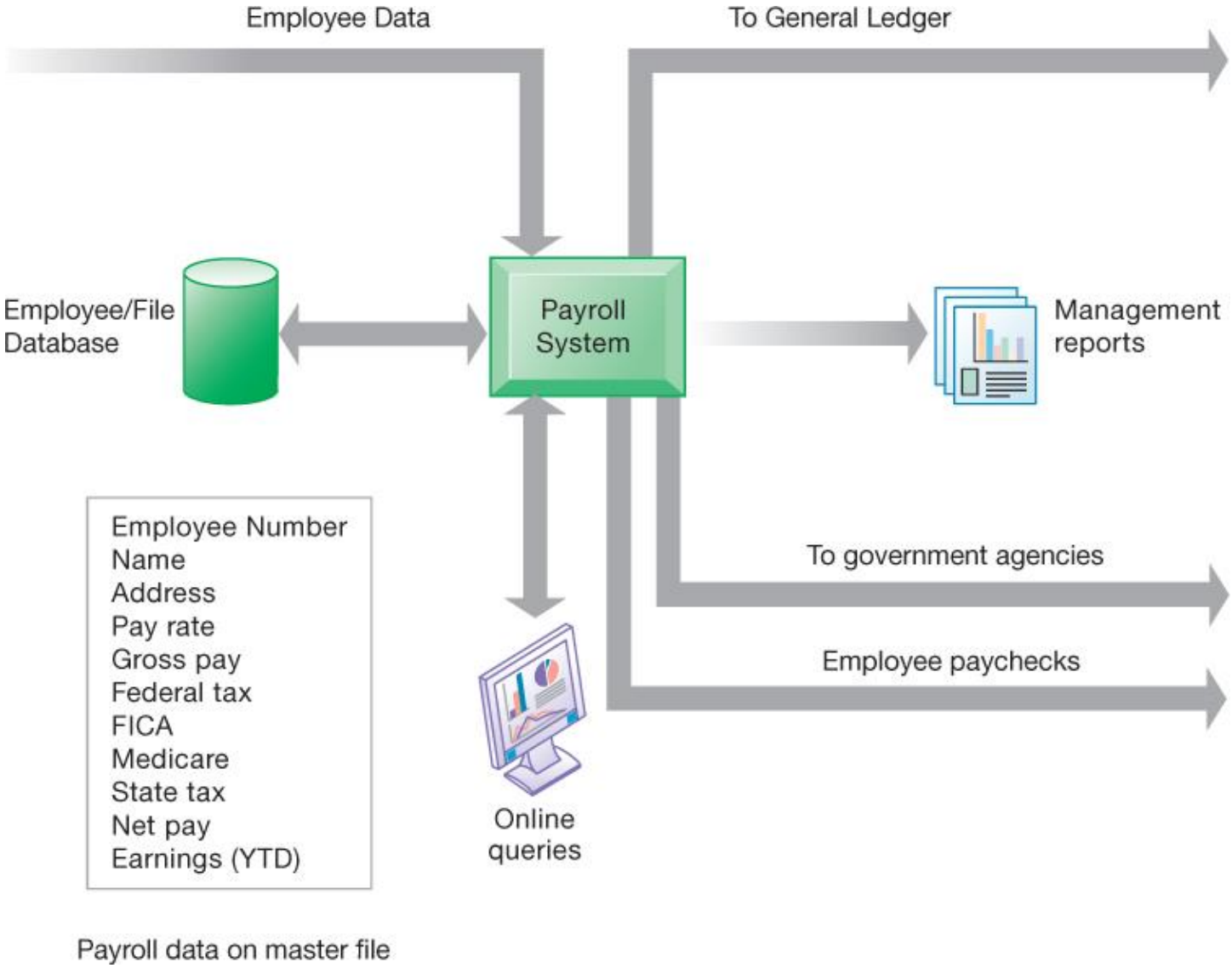
Type of System	Information Inputs	Information Outputs	Main users
Transaction Processing Systems (TPS)	• Transactions • Daily events	• Detailed reports • Lists • Summaries	• Operations personnel • First-line supervisors
Management Information Systems (MIS)	• Summary transaction data • High-volume data • Simple models	• Summary reports • Exception reports	• Middle managers
Decision Support Systems (DSS)	• Data optimized for analysis • Analytical models • Data analysis tools	• Interactive analyses • Simulations	• Professionals • Staff managers
Executive Support Systems (ESS)	• Aggregated internal and external data	• Projections • Responses to queries	• Senior managers

Transaction Processing System (TPS)

- **A computerized system that performs and records the daily routine transactions necessary to conduct business**
  - e.g., sales order entry, hotel reservations, payroll, employee record keeping, shipping
  - Generate critical data for other systems and business functions, integrating operations across the organization.
  - Essential for monitoring internal operations and managing external relationships.
  - Answer routine queries and monitor the transaction flow within the organization, ensuring information is accessible, up-to-date, and accurate.
- **Critical Business Infrastructure:**
  - Vital to a business's core functions; a brief failure can significantly impact the business and its interconnected networks.

A Payroll TPS

- **Interacts with employee database and generate management reports**



Business Intelligence

- **Systems for business intelligence**
  - BI: Data and software tools for organizing, analyzing, and providing access to data to help managers and other users make ***informed decisions***

1. **Management information systems (MIS)**
2. **Decision support systems (DSS)**
3. **Executive support systems (ESS)**

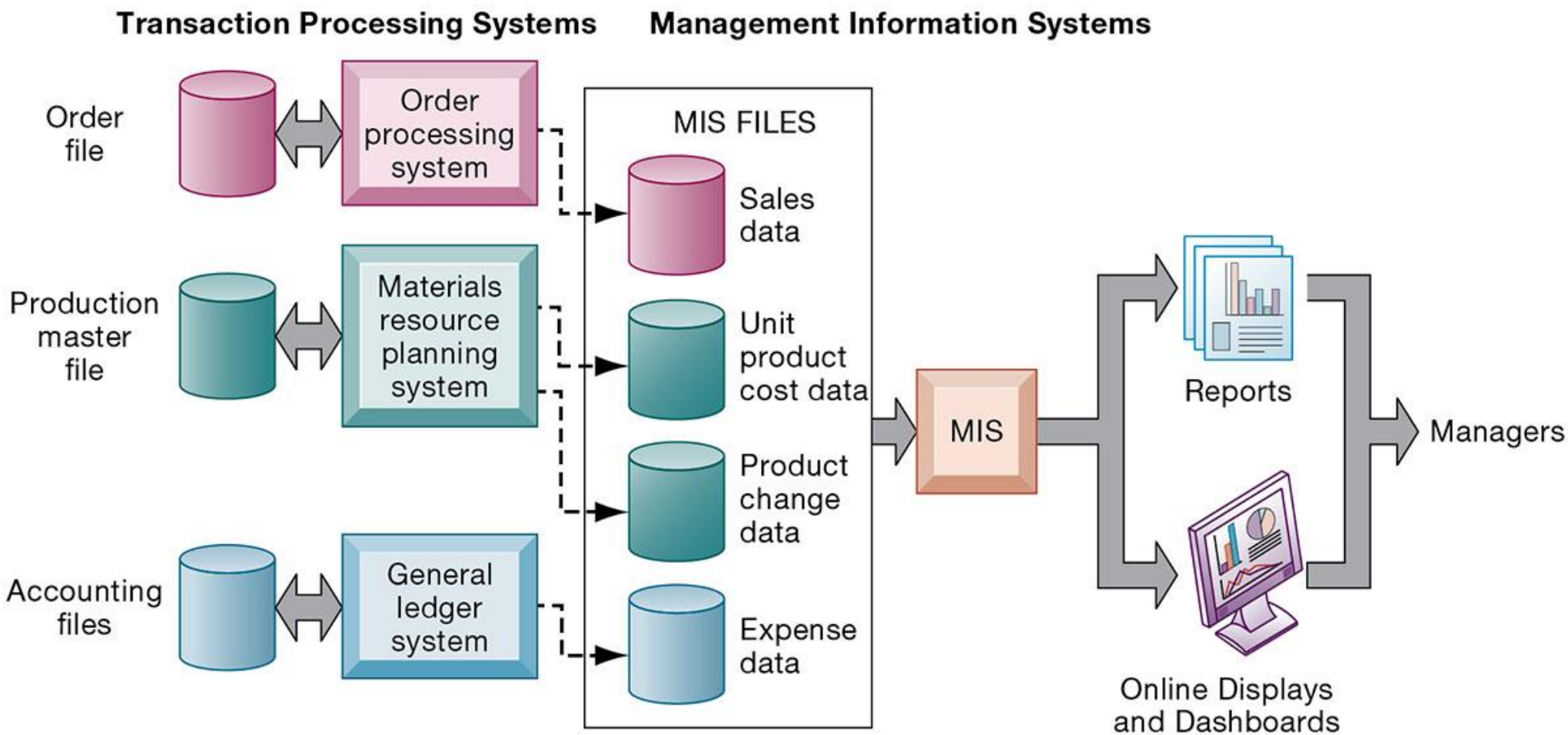
Management Information Systems (MIS)

1

- **A field of study as well as a specific category of information systems serving middle management.**
- **Provides reports on the organization's current performance to monitor, control, and forecast future performance.**
- ***Regularly summarize and report* on the company's basic operations using data supplied by TPS.**
- ***Routine Question Handling*: answer predefined questions with established procedures, such as tracking inventory usage or comparing sales figures against targets.**
- ***Limited Flexibility and Analytical Depth*: minimal analytical capabilities, focusing on simple data aggregation like summaries and comparisons rather than advanced statistical analysis.**

Example: Sample MIS Report

- **Provides reports and online displays and dashboard that helps managers**



Example: Sample MIS Report

1

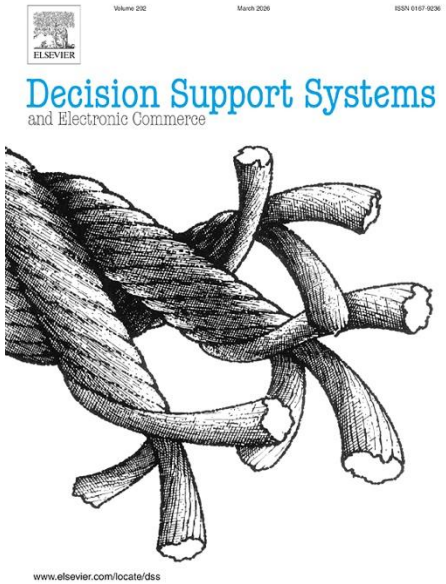
- **Consolidated Consumer Products Corporation Sales by Product and Sales Region: 2020**

PRODUCT CODE	PRODUCT DESCRIPTION	SALES REGION	ACTUAL SALES	PLANNED	ACTUAL versus PLANNED
4469	Carpet Cleaner	Northeast	4,066,700	4,800,000	0.85
		South	3,778,112	3,750,000	1.01
		Midwest	4,867,001	4,600,000	1.06
		West	4,003,440	4,400,000	0.91
	TOTAL		16,715,253	17,550,000	0.95
5674	Room Freshener	Northeast	3,676,700	3,900,000	0.94
		South	5,608,112	4,700,000	1.19
		Midwest	4,711,001	4,200,000	1.12
		West	4,563,440	4,900,000	0.93
	TOTAL		18,559,253	17,700,000	1.05

Decision Support Systems (DSS)

2

- **Focus on unique or rapidly changing problems, for which the procedure to a solution may not be fully predefined in advance**
  - Analyze potential impacts on business operations, such as changes in sales or investment returns, under various scenarios.
  - Use *internal* data from TPS and MIS with *external* information like market prices and competitor data
  - Utilized by managers and business analysts who want to use *sophisticated* analytics and models to analyze data.
- **Model-driven DSS vs. Data-driven DSS**
  - e.g. Voyage-estimating systems vs. Intrawest's marketing analysis systems



Model-Driven DSS vs. Data-Driven DSS

2

- **Theoretical models and simulations vs. practical data analysis**
  - M: Utilizes *sophisticated mathematical* models, simulations, and analytical techniques
  - D: Utilizes *extensive data repositories* and employ data management and processing tools to extract insights.
- **Explore the outcomes of different *scenarios* through models vs. reveal insights within data that can guide decision-making**
  - M: Problem Solving - targets unstructured or semi-structured management problems where standard procedural solutions may not exist.
  - D: Finding insights - uncover patterns, trends, and insights from raw data, including historical data, transaction data, or real-time data streams.
- **Different application areas**
  - M: financial forecasting, supply chain logistics, and strategic planning.
  - D: marketing analysis, customer behavior studies, and operational performance

Example of Model-Driven DSS

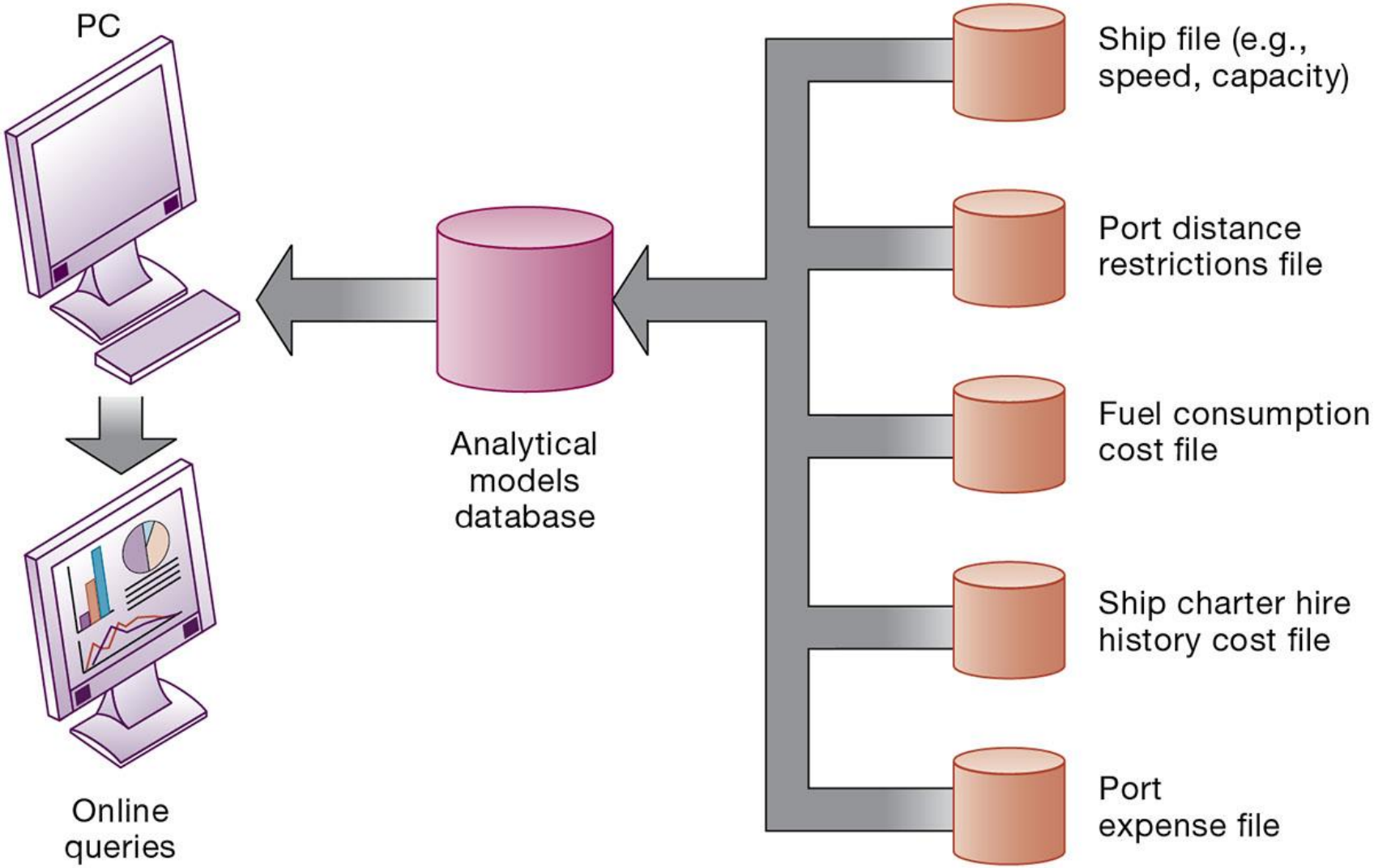
2

- **Voyage-Estimating DSS:** Designed for a global shipping company to aid in *voyage planning* and profitability analysis.
- **Capabilities:**
  - ***Financial Analysis***: Computes **costs** associated with ship operation (fuel, labor, capital), freight rates, and port expenses.
  - ***Technical Analysis***: Assesses ship cargo capacity, speed, distance to ports, fuel and water consumption, and loading patterns.
- **Determines the most cost-effective vessel assignment, optimal vessel speed for profit maximization**
- **Operates on a desktop PC, providing a system of menus that makes it easy for users to enter data or obtain information**

Voyage-Estimating Decision-Support System

2

- **Example of Model-Driven DSS**



Example of Data-Driven DSS

2

- **DSS of Vail Resorts:** Analyze customer data to evaluate customer value, revenue potential, and loyalty
- **Capabilities:**
  - ***Accumulate*** extensive customer data from various sources, including call centers, lodging, ski schools, and rental stores, and ***analyze*** the data to find insights
- **Use insights to refine and target *marketing strategies*, enhancing customer engagement and resource allocation decision-making.**

Image Source: Vail Resorts

Executive Support Systems (ESS)

3

- **Play a crucial role in assisting senior management with strategic decision-making**
  - Supports *complex, nonroutine* decisions that require judgment, evaluation, and insight, lacking a predefined procedure for resolution.
  - Offers an accessible interface for senior managers, presenting data and graphs
  - Information is often accessible through a web portal, offering an integrated and personalized view of business content.
  - Integrates *external and internal* data sources to provide a comprehensive view encompassing market conditions and internal performance metrics.
- **Critical Data Presentation:** Filters, compresses, and tracks critical data, focusing on information most relevant to senior managers
- **Advanced Analytics:** increasingly equipped with BI analytics features, such as trend analysis, forecasting, and detailed data exploration

Example of ESS

3

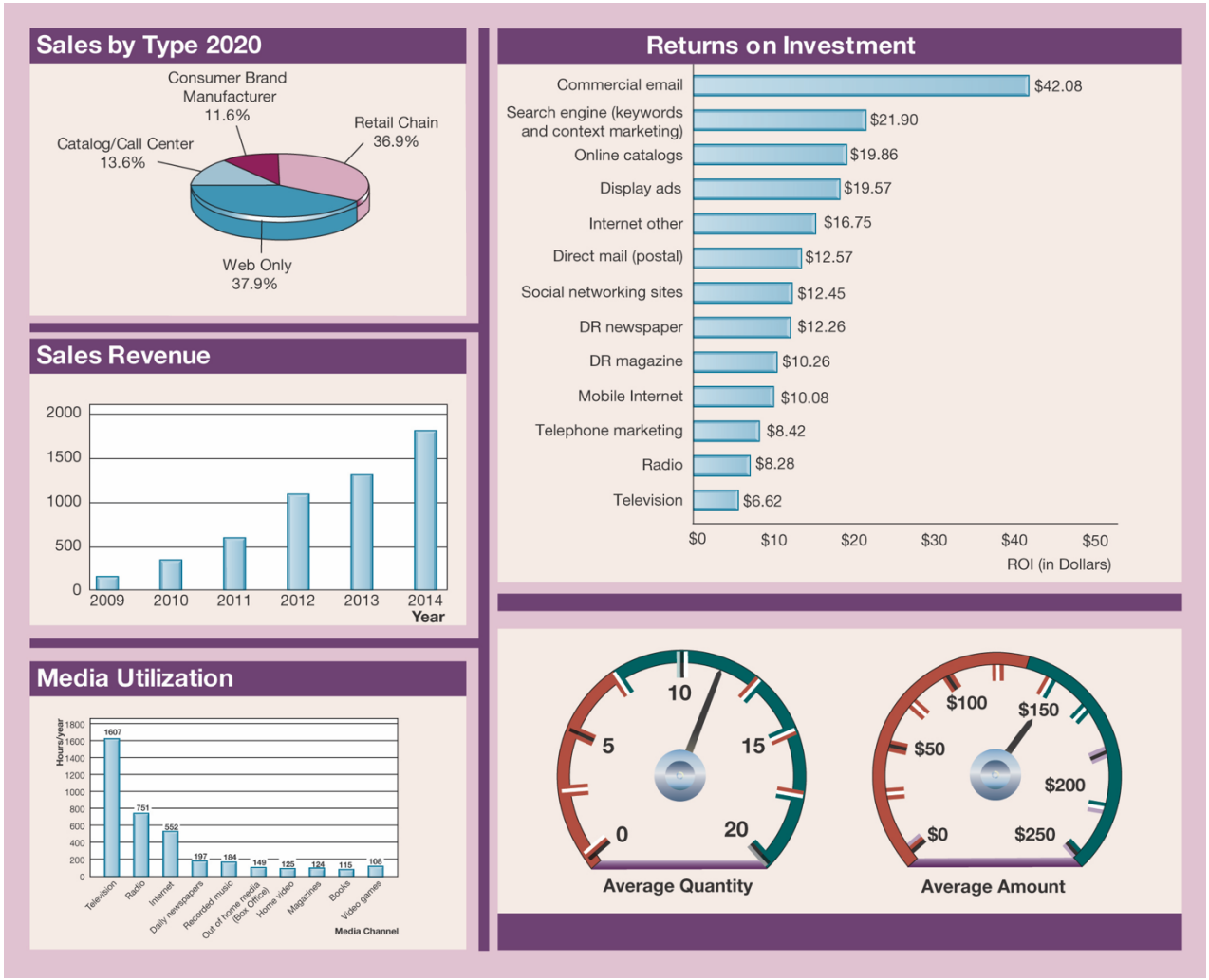
- **Valero's Refining Dashboard:** Used by the *COO and plant managers* to monitor real-time operational data
  - Reliability, inventory, safety, and energy consumption

- **Enables *performance review* of each refinery against the firm's production plan, with capabilities to analyze data from the executive level down to individual operators.**

- **Serves as an example of a digital dashboard**
  - A tool that consolidates ***KPIs*** on a single screen for effective management oversight.

Digital Dashboard Example

- **Delivers comprehensive and accurate information for decision-making, often using a single screen**



The graphical overview of _____ helps managers quickly spot areas that need attention.

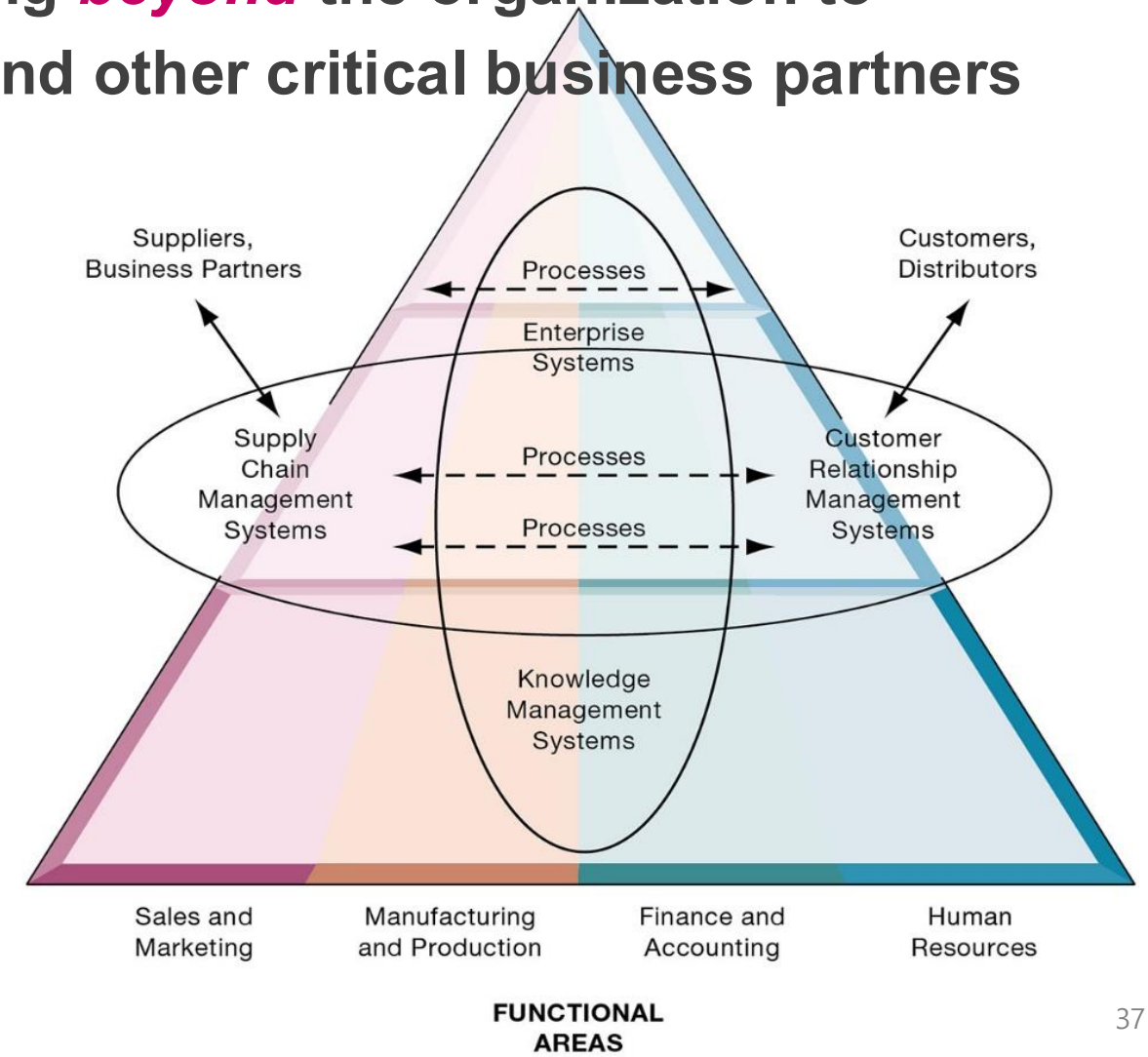
Enterprise Applications

How to manage all the information in these different systems?
how to coordinate their work in these different systems?
How to maintain so many different systems?

- **Integrating different systems within a company is a *major challenge***
- **Solution: Enterprise Applications (i.e., Systems for linking the enterprise)**
  - Designed to span across functional areas, focusing on executing business processes ***throughout the entire firm*** and involving all levels of management.
  - Aim to make businesses ***more flexible and productive*** by closely coordinating business processes and integrating various processes for efficient resource management and improved customer service.

Enterprise Application Architecture

- **Encompasses processes spanning *the entire* organization and, in some cases, extending *beyond* the organization to customers, suppliers, and other critical business partners**



Types of Enterprise Applications

1. **Enterprise Systems (Enterprise Resource Planning - ERP):**
   Integrate core business processes in ***finance, HR, manufacturing, sales, etc., into a single system*** to streamline operations and data management across the organization

2. **Supply Chain Management (SCM) Systems:**
   Manage the ***flow of goods***, information, and finances from supplier to consumer, optimizing the supply chain operations.

3. **Customer Relationship Management (CRM) Systems**:
   Focus on managing a company's ***interactions with current and potential customers*** to improve relationships and increase sales.

4. **Knowledge Management Systems (KMS):**
   Facilitate the organization, storage, retrieval, and dissemination of ***knowledge*** within the organization.

ERP

1

- **Integrate business processes in different functions into a single software system**
- ***Unified* Data Repository:** centralize data in one comprehensive database, making it accessible to different parts of the business and helping to maintain data consistency and accuracy
- **Streamlined Operations:** automates the ***flow of information*** across the organization
  - e.g., Customer order → Shipping & fulfillment → Inventory replenishment → Billing → Customer service updates
- ***Informed Decision-Making***
  - Enable managers to make more precise and timely decisions and to perform daily operational management and longer-term strategic planning.

SCM

2

- **Objective: to get the *right* amount of their products from their source for consumption in the *least* amount of time and at the *lowest* cost**
- **Help manage relationships with their suppliers, purchasing firms, distributors, and logistics companies**
  - Help them share information about orders, production, inventory levels, and delivery of products and services.
  - Help them source, produce, and deliver goods and services *efficiently*
- **Profitability Improvement**
  - By lowering the costs of moving and making products
  - By enabling managers to make better decisions about how to organize and schedule sourcing, production, and distribution
- **Interorganizational System: automate the flow of information across organizational boundaries**

CRM

3

- **Objective: to provide information to coordinate all business processes dealing with *customers* in sales, marketing, and service**
- **Helps manage relationships with customers.**
- **By providing a centralized platform for information about customer interactions, CRM systems aim to increase revenue, customer satisfaction, and customer retention.**
  - ***Targeting Profitable Customers***: identify, attract, and retain the most profitable customers by leveraging data to understand customer behaviors and preferences.
  - ***Enhancing Service and Sales***: provide better service to existing customers and effectively increase sales through targeted marketing and personalized service offerings.

KMS

4

- **Strategic Use of Firm *Knowledge***
  - Knowledge is a key differentiator and a challenge for competitors to replicate
- **Enable organizations to better manage processes for capturing and applying knowledge and expertise**
  - Collect all relevant knowledge and experience in the firm and make it available wherever and whenever needed to improve business processes and management decisions.
- **Beyond internal knowledge, KMS also facilitates access to *external* sources of knowledge, enriching the firm's understanding and capabilities with outside insights**

Intranets and Extranets

- **Intranets and extranets are key digital tools that businesses use to *enhance information flow and integration* both within the organization and with external partners**
- **Intranets: Internal company websites accessible exclusively to employees**
  - Serve as a central platform for sharing internal information, resources, and communication
- **Extranets: Extensions of intranets that are accessible to authorized *external users* such as vendors, suppliers, and sometimes customers**
  - Facilitate secure communication and collaboration with external stakeholders, streamlining SCM, customer service, and partner relations.

2.3

Why are systems for collaboration and social business so important?

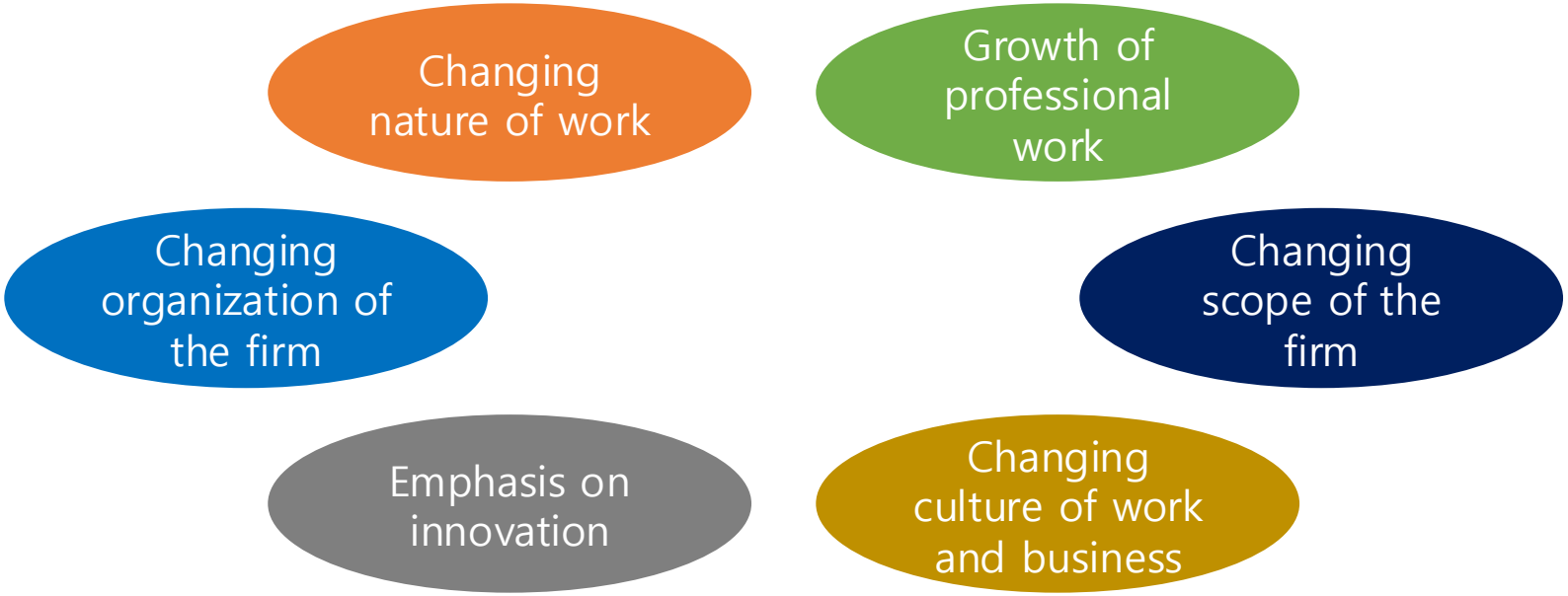
&

What technologies do they use?

What Is Collaboration?

- **Collaboration is working with others *to achieve shared and explicit goals***
  - Short-lived vs. long term
  - One-to-one vs. many-to-many
  - Informal vs. formal

Growing Importance of Collaboration

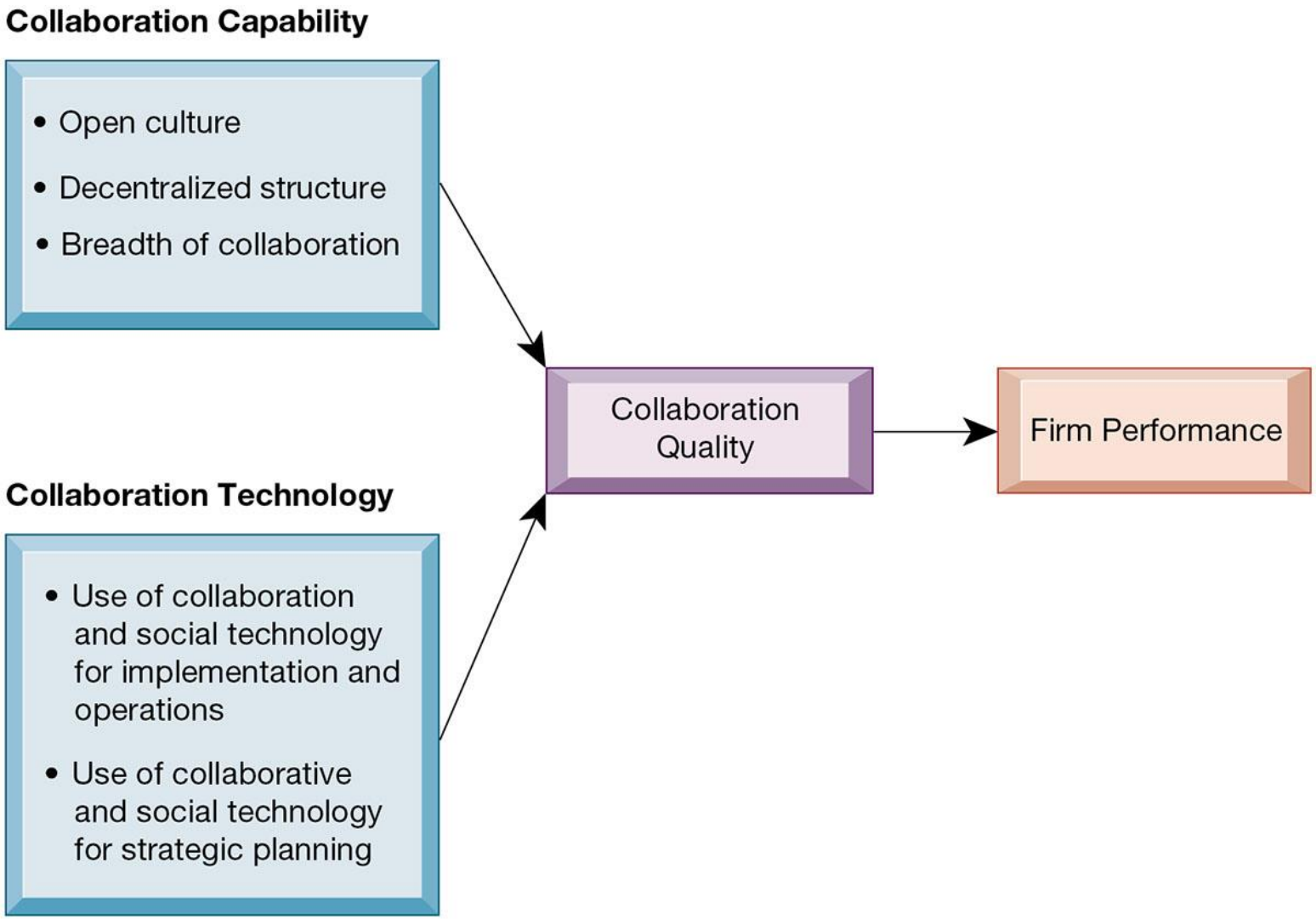


Social Business

- **Social business uses *social networking tools* to enable open interaction and communication inside and outside the firm.**
- **Focuses on ongoing conversations among customers, employees, suppliers, and partners**
  - By participating in these conversations, firms can build stronger relationships and increase emotional engagement with stakeholders.
- **Social business increases *transparency* by allowing information, opinions, and feedback to flow directly across organizational boundaries.**
- **This *direct sharing of knowledge* helps organizations improve efficiency, innovate faster, and make quicker, better-informed decisions.**

Requirements for Collaboration

- **Collaboration capability and collaboration technology matter**



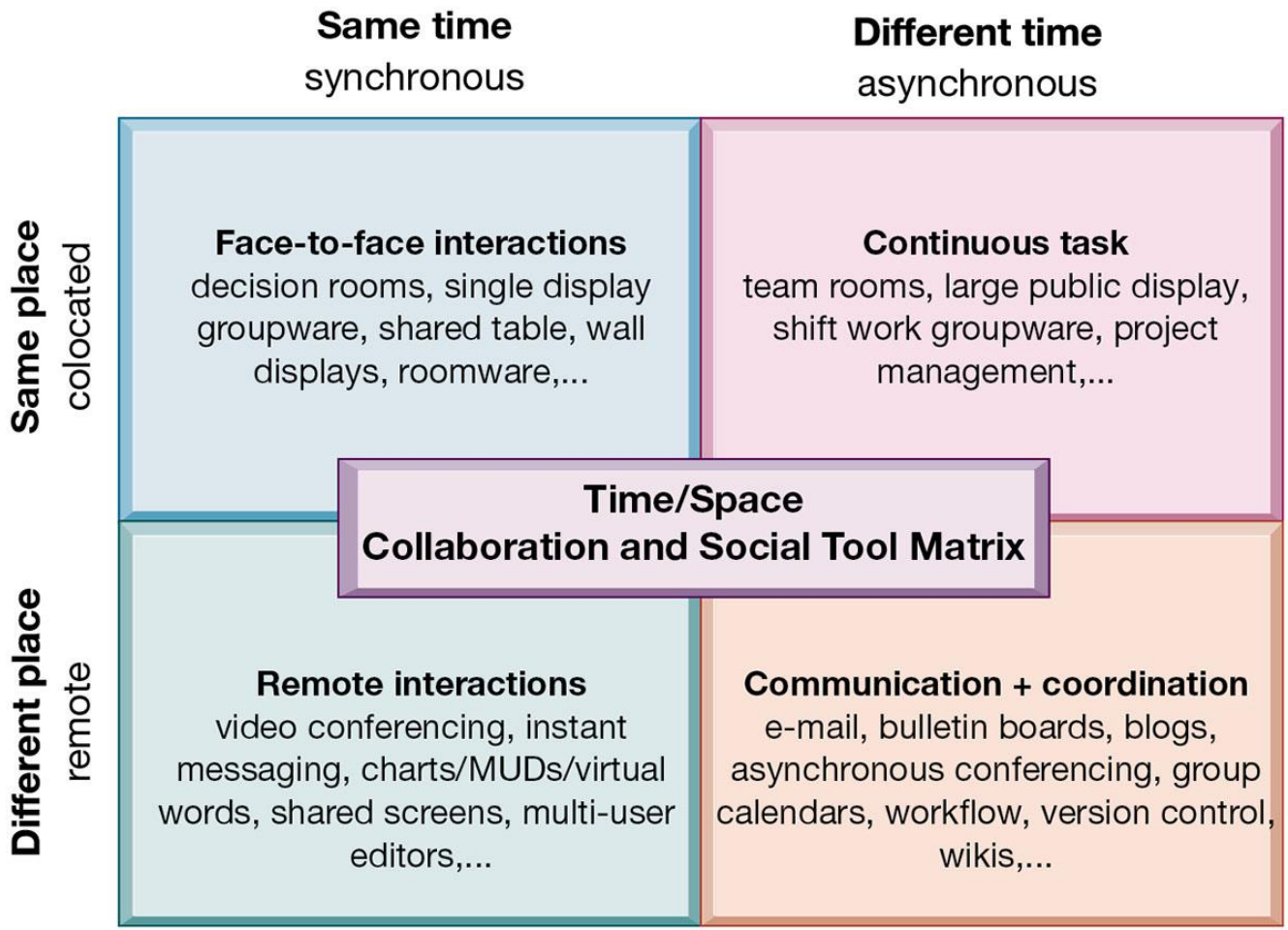
Tools and Technologies for Collaboration and Social Biz

- **E-mail and instant messaging (IM)**
- **Wikis**
- **Virtual worlds**
- **Collaboration and social business platforms**
  - Virtual meeting systems (telepresence)
  - Cloud collaboration services (Google Drive)
  - Enterprise social networking tools
  - Etc.

Image Source: YouTube

Evaluating and Selecting Collaboration and Social Software Tools

- **Time Dimension: Synchronous vs. Asynchronous**
- **Space Dimension: Collocated vs. remote**



2.4

What is the role of the information systems in a business?

The Information Systems Department

- **The organizational unit responsible for IT services**
  - Maintain the hardware, software, data storage, and networks
- **Often headed by chief information officer (CIO)**
  - Oversee the use of information technology in the firm
  - Expected to have a strong business background and information systems expertise and to play a key role in integrating technology into the firm's business strategy
- **Consists of specialists, such as programmers, systems analysts, project leaders, and information systems managers**
- **End users**
  - Representatives of departments outside of the information systems group for whom applications are developed

Organizing the Information Systems Function

- **Organizing the Information Systems Function**
  - Small company: ***one employee*** who is responsible for keeping its networks and applications running
  - Larger companies: a separate information systems ***department***

- **IT Governance**
  - Strategies and policies for using IT in the organization
  - Decision rights and framework for accountability
  - Organization of information systems function
    - Centralized, decentralized, and so on

Discussion

- **Which type of enterprise system (ERP, SCM, CRM, or KMS) do you think would benefit the most from AI support, and why?**

Image Source: Salesforce

Key Takeaways

- Business processes are structured sets of activities that organizations use to produce goods or services, and information systems help execute, monitor, and improve these processes.
- Information systems support different management levels by providing the right type of information for operational, managerial, and strategic decision-making.
- Collaboration and social business systems help employees and external partners share information, coordinate work, and innovate more effectively.
- The information systems function plays a critical role in aligning technology with business goals and ensuring systems deliver value to the organization.

Next Week...

- **Next week, we will cover:**
  - [Chapter 3] Information Systems, Organizations, and Strategy

Q & A